

HOMEWORK 11
ELEMENTARY LINEAR ALGEBRA
(MATH 2250-03), SPRING, 2019

CHECK: 05/08/2019

SECTION 6.4 - THE GRAM-SCHMIDT PROCESS

1. Let $v_1 = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 8 \\ 5 \\ 6 \end{bmatrix}$, and write $W = \text{span}\{v_1, v_2\}$. Use the Gram-Schmidt process to find an orthogonal basis for W .

Recall that the Gram-Schmidt Process (Theorem 11 of Section 6.4 in your textbook) states that if $\{x_1, \dots, x_p\}$ is a basis for a nonzero subspace W of $\mathbb{R}^n$, then the set $\{u_1, \dots, u_p\}$ is an orthogonal basis for W , where

$$\begin{aligned} u_1 &= x_1 \\ u_2 &= x_2 - \frac{x_2 \cdot u_1}{u_1 \cdot u_1} u_1 \\ u_3 &= x_3 - \frac{x_3 \cdot u_1}{u_1 \cdot u_1} u_1 - \frac{x_3 \cdot u_2}{u_2 \cdot u_2} u_2 \\ &\vdots \\ u_p &= x_p - \frac{x_p \cdot u_1}{u_1 \cdot u_1} u_1 - \frac{x_p \cdot u_2}{u_2 \cdot u_2} u_2 - \dots - \frac{x_p \cdot u_{p-1}}{u_{p-1} \cdot u_{p-1}} u_{p-1}. \end{aligned}$$

So, with this in mind, note that the set $\{v_1, v_2\}$ given in the statement of the problem is a basis for the subspace $W = \text{span}\{v_1, v_2\}$, as $\{v_1, v_2\}$ is a linearly independent set. So, since W is a nonzero subspace of $\mathbb{R}^3$ and $\{v_1, v_2\}$ is a basis for W , we may use the Gram-Schmidt process on $\{v_1, v_2\}$ to produce an orthogonal basis $\{u_1, u_2\}$ of W by setting:

$$u_1 = v_1 = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} \quad \text{and} \quad u_2 = v_2 - \frac{v_2 \cdot u_1}{u_1 \cdot u_1} u_1 = v_2 - \frac{9}{5} v_1 = \begin{bmatrix} 13/5 \\ 5 \\ 39/5 \end{bmatrix}.$$

While the above is enough to correctly answer this question on your homework or an exam, let's take a moment to check that $\{u_1, u_2\}$ is indeed an orthogonal set:

$$u_1 \cdot u_2 = (3)(13/5) + (0)(5) + (-1)(39/5) = 39/5 - 39/5 = 0.$$

So, we're good there. For fun, let's also verify that $\{u_1, u_2\}$ spans W . To do so, notice that, as $u_1 = v_1$ and $u_2 = v_2 - \frac{v_2 \cdot u_1}{u_1 \cdot u_1} u_1$, this then gives $v_2 = u_2 + \frac{v_2 \cdot u_1}{u_1 \cdot u_1} u_1$. So, as $W = \text{span}\{v_1, v_2\}$, it follows that if $y \in W$, then there are constants $c_1, c_2 \in \mathbb{R}$ such that $y = c_1 v_1 + c_2 v_2$. But, using the above, this then yields that

$$y = c_1 u_1 + c_2 \left(u_2 + \frac{v_2 \cdot u_1}{u_1 \cdot u_1} u_1 \right) = \left(c_1 + c_2 \frac{v_2 \cdot u_1}{u_1 \cdot u_1} \right) u_1 + c_2 u_2.$$

So, there are constants d_1 and d_2 such that $y = d_1 u_1 + d_2 u_2$, and $\text{span}\{v_1, v_2\} = \text{span}\{u_1, u_2\}$. Therefore, as the Gram-Schmidt process assures, $\{u_1, u_2\}$ is indeed an orthogonal basis for W .

2. Let $A = \begin{bmatrix} 3 & -5 & 1 \\ 1 & 1 & 1 \\ -1 & 5 & -2 \\ 3 & -7 & 8 \end{bmatrix}$. Find an orthogonal basis for $\text{Col}(A)$.

To begin, let's notice that routine calculations yield that A is row-equivalent to the matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$. So, since this tells us that the first, second, and third columns of A are pivot columns, the first, second, and third columns of A form a basis for $\text{Col}(A)$. That is, the columns of A are linearly independent and the set $\{a_1, a_2, a_3\}$ is a basis for $\text{Col}(A)$, where

$$a_1 = \begin{bmatrix} 3 \\ 1 \\ -1 \\ 3 \end{bmatrix}, \quad a_2 = \begin{bmatrix} -5 \\ 1 \\ 5 \\ -7 \end{bmatrix}, \quad a_3 = \begin{bmatrix} 1 \\ 1 \\ -2 \\ 8 \end{bmatrix}.$$

Since $\{a_1, a_2, a_3\}$ is a basis for the nonzero subspace $\text{Col}(A)$ of $\mathbb{R}^4$, we may use the Gram-Schmidt process on $\{a_1, a_2, a_3\}$ to produce an orthogonal basis $\{u_1, u_2, u_3\}$ for $\text{Col}(A)$ by setting

$$u_1 = a_1 = \begin{bmatrix} 3 \\ 1 \\ -1 \\ 3 \end{bmatrix}$$

$$u_2 = a_2 - \frac{a_2 \cdot u_1}{u_1 \cdot u_1} u_1 = a_2 + 2u_1 = \begin{bmatrix} 1 \\ 3 \\ 3 \\ -1 \end{bmatrix}$$

$$u_3 = a_3 - \frac{a_3 \cdot u_1}{u_1 \cdot u_1} u_1 - \frac{a_3 \cdot u_2}{u_2 \cdot u_2} u_2 = a_3 - \frac{3}{2}u_1 + \frac{1}{2}u_2 = \begin{bmatrix} -3 \\ 1 \\ 1 \\ 3 \end{bmatrix}.$$

SECTION 6.5 - LEAST-SQUARES PROBLEMS

3. Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 3 \\ 8 \\ 2 \end{bmatrix}$. Describe all least-squares solutions to the equation $Ax = b$.

Recall that Theorem 13 of Section 6.5 in your textbook states that the set of least-squares solutions of $Ax = b$ coincides with the nonempty set of solutions of the normal equation $A^T Ax = A^T b$. So, notice that

$$A^T A = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 2 \end{bmatrix}, \quad A^T b = \begin{bmatrix} 14 \\ 4 \\ 10 \end{bmatrix}.$$

Correspondingly, we can form an augmented matrix for the inhomogeneous linear system $A^T Ax = A^T b$ and row reduce as follows:

$$\left[\begin{array}{cccc} 4 & 2 & 2 & 14 \\ 2 & 2 & 0 & 4 \\ 2 & 0 & 2 & 10 \end{array} \right] \xrightarrow{\text{Row Reduce}} \left[\begin{array}{cccc} 1 & 0 & 1 & 5 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

The general solution to the row-equivalent system is then given by $x_1 = -x_3 - 5x_4$ and $x_2 = x_3 + 3x_4$, where x_3 and x_4 are free variables. In parametric vector form, this says that the solutions to the system $A^T Ax = A^T b$ are given by

$$x = \begin{bmatrix} -x_3 - 5x_4 \\ x_3 + 3x_4 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -5 \\ 3 \\ 0 \\ 1 \end{bmatrix}.$$

Since we know by Theorem 13 that the solutions to $A^T Ax = A^T b$ coincide with the least-squares solutions to $Ax = b$, this yields that the above is a least squares solution to $Ax = b$ for any choice of x_3 and x_4 . Consequently, we may conclude that the set of all the least-squares solutions to the equation $Ax = b$ is given by

$$\text{span} \left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ 3 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

4. Let A be an $m \times n$ matrix such that $A^T A$ is invertible. Show that the columns of A are linearly independent.

(WARNING: Since A may not be a square matrix, you may **not** assume that A is invertible!)

Let A be an $m \times n$ matrix such that $A^T A$ is invertible. We want to show that the columns of A are linearly independent.

To do so, let us note that since $A^T A$ is invertible, the homogeneous system $A^T A x = 0$ has only the trivial solution $x = 0$. Consequentially, since any vector $x \in \text{Nul}(A)$ satisfies $Ax = 0$, it must be the case that $\text{Nul}(A) = \{0\}$, since the inclusion of any nonzero vector y in $\text{Nul}(A)$ would then produce a nontrivial solution to $A^T A x = 0$. That is, if there existed a vector $y \neq 0$ belonging to $\text{Nul}(A)$, we would have $A^T A y = A^T(0) = 0$ for some nonzero vector y , contradicting the invertibility of $A^T A$.

So, writing $A = \begin{bmatrix} a_1 & \dots & a_n \end{bmatrix}$, where $a_1, \dots, a_n \in \mathbb{R}^m$ are the columns of A , let us recall that

$$Ax = \begin{bmatrix} a_1 & \dots & a_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1 a_1 + \dots x_n a_n.$$

Since $Ax = 0$ has only the trivial solution $x = 0$, this then says that $x_1 a_1 + \dots x_n a_n = 0$ if and only if $x_1 = \dots = x_n = 0$. Since this is precisely the definition of linear independence, we may conclude that the columns of A must be linearly independent whenever $A^T A$ is invertible.